

Adroit Technologies

Protocol Driver

Name	Mitsubishi MX Components
DLL	MITMXACT



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Contents

1. Overview	4
1.1. Upgrading.....	4
2. Wiring	5
3. Device Configuration	6
3.1. Device Configuration	6
3.1.1. Port Selection	6
3.1.2. PLC Details.....	6
4. Supported Addresses	7
4.1. Mitsubishi Q Series (System Q) PLC.....	7
4.2. Mitsubishi QnA Series PLC.....	8
4.3. Mitsubishi L Series PLC	9
4.4. Mitsubishi iQ-R Series PLC	10
4.5. Mitsubishi A Series PLC.....	11
4.6. Mitsubishi iQ-F Series PLC.....	12
4.7. Omitted	13
4.8. Non-bit Device Writes.....	13
4.9. Bit Devices:	13
4.10. Word Devices:	14
4.11. PLC Software	15
4.11.1. Installing MX Components	15
4.11.2. Configuring a new communication channel using the utility.....	15
5. Protocol Definition and Considerations	18
6. General Notes and Troubleshooting	19
6.1. Contact Details.....	19
6.2. Adroit Technologies Knowledge Base.....	19
6.3. Troubleshooting	19
6.3.1. Troubleshooting Steps.....	19
6.3.2. Error Message Reference.....	20
7. Revisions.....	20
7.1. Document Revision.....	20
7.2. DLL Revision	20

1. Overview

The Mitsubishi MX Components driver makes use of the Mitsubishi MX Components ActiveX Control supplied by Mitsubishi. The MX Components installation and License can be purchased from Mitsubishi.

Please ensure MX Components is installed for this driver to function correctly. The driver also uses the Mitsubishi Communication Setup Utility that gets installed with MX Components for configuring communications with PLCs.



WARNING

The driver only supports MX Components 4 or later.

1.1. Upgrading

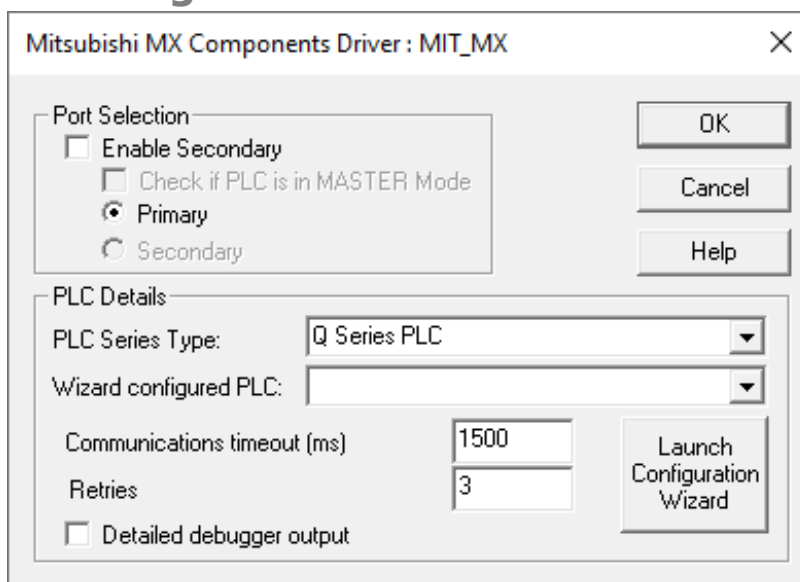
If the MITQJE71 driver is being used, remove all the old Device Agents and create them again with the same names using this driver. Restart the Agent Server and all your scanning will remain, but on the new MX Components driver.

2. Wiring

Standard Ethernet.

3. Device Configuration

3.1. Device Configuration



3.1.1. Port Selection

Enable secondary – Turn this option to have the driver maintain 2 channels of communication, the radio buttons **Primary** and **Secondary** can then be used to toggle between configurations.

Check PLC is in MASTER Mode – When two PLC's are configured for redundancy, a check is performed for a PLC Bit (SM1516) to establish which PLC is in MASTER mode before doing a WRITE to the PLC. This functionality can be disabled by deselecting the box, causing the driver will not check for this bit in the PLC and will just do the write to the PLC it is connected to.

3.1.2. PLC Details

PLC Series type – Select the PLC Series type namely FX Series, Q Series, L Series, A series, R Series or QnA series PLC.

Wizard Configured PLC – Select the relevant communication configuration that was done by the Mitsubishi communications setup utility.

Launch Wizard Configuration – Click to open the Mitsubishi communication setup utility.

Communications timeout – The interval in milliseconds before the driver assumes the PLC is not responding if not response is received, the driver will reset and open the connection. As in the case with a non-fatal error exception response, the driver will re-attempt the transaction three times. The number of retries is not settable.

Detailed debugger output – Turning this option on forces the driver to provide detailed information of activity on the device's driver diagnostic datascope screen. Use this option when troubleshooting problems involving the watchdog/heartbeat, or specific tags not working.

Retries – The amount of retries for a specific scan task before returning failed.

Launch Configuration Wizard – Launches the MX Components configuration wizard. Ensure the correct version of MX Components is installed.

4. Supported Addresses

4.1. Mitsubishi Q Series (System Q) PLC

Mitsubishi Q series (System Q) PLC data (device) types supported in the Adroit Protocol.

Device	Prefix	Max Range	PLC Data Type	Boolean	Integer	Real	String
Inputs	X	0x1FFF	discrete	X 0			
			Int 16		X0	X0	
			Word 16		X0 W	X0 W	
			dword 32		X0 D	X0 D	
			Long 32		X0 L	X0 L	
			Float 32		X0 F	X0 F	
Output relay	Y	0x1FFF	discrete	Y 0			
Internal relay	M	8191	discrete	M 0			
Latch relay	L	8191	discrete	L 0			
Annunciator	F	2047	discrete	F 0			
Link relay	B	0x1FFF	discrete	B 0			
Special link relay	SB	0x7FF	discrete	SB 0			
Edge relay	V	2047	discrete	V 0			
Data register	D	12287	bit	D0.0			D 0 Sn
			Int 16		D0	D0	
			Word 16		D0 W	D0 W	
			dword 32		D0 D	D0 D	
			Long 32		D0 L	D0 L	
			Float 32		D0 F	D0 F	
Link register	W	0x1FFF	Int 16		W 0	W 0	
Timer (current value)	T or TN*	2047	Int 16		TN 0	TN 0	
Retentive timer (current value)	ST	2047	Int 16		ST 0	ST 0	
Counter (current value)	C or CN*	1023	Int 16		CT 0	CN 0	
Special link reg.	SW	0x7FF	Int 16		SW 0	SW 0	
File register	R	32767	bit	R0.0			R 0 Sn
			Int 16		R0	R0	
			Word 16		R0 W	R0 W	
			dword 32		R0 D	R0 D	
			Long 32		R0 L	R0 L	
			Float 32		R0 F	R0 F	
Extended File register	ZR	4184063	bit	ZR0.0			
			Int 16		ZR0	ZR0	
			Word 16		ZR0 W	ZR0 W	
			dword 32		ZR0 D	ZR0 D	
			Long 32		ZR0 L	ZR0 L	
			Float 32		ZR0 F	ZR0 F	

4.2. Mitsubishi QnA Series PLC

Mitsubishi QnA series PLC data (device) types supported in the Adroit Protocol.

Device	Prefix	Max Range	PLC Data Type	Boolean	Integer	Real	String
Inputs	X	0x1FFF	discrete	X 0			
			Int 16		X0	X0	
			Word 16		X0 W	X0 W	
			dword 32		X0 D	X0 D	
			Long 32		X0 L	X0 L	
			Float 32		X0 F	X0 F	
Output relay	Y	0x1FFF	discrete	Y 0			
Internal relay	M	8191	discrete	M 0			
Latch relay	L	8191	discrete	L 0			
Annunciator	F	2047	discrete	F 0			
Link relay	B	0x1FFF	discrete	B 0			
Special link relay	SB	0x7FF	discrete	SB 0			
Edge relay	V	2047	discrete	V 0			
Data register	D	8191	bit	D0.0			D 0 Sn
			Int 16		D0	D0	
			Word 16		D0 W	D0 W	
			dword 32		D0 D	D0 D	
			Long 32		D0 L	D0 L	
			Float 32		D0 F	D0 F	
Link register	W	0x1FFF	Int 16		W 0	W 0	
Timer (current value)	T or TN*	2047	Int 16		TN 0	TN 0	
Retentive timer (current value)	ST	2047	Int 16		ST 0	ST 0	
Counter (current value)	C or CN*	1023	Int 16		CT 0	CN 0	
Special link reg.	SW	0x7FF	Int 16		SW 0	SW 0	
File register	R	32767	bit	R0.0			R 0 Sn
			Int 16		R0	R0	
			Word 16		R0 W	R0 W	
			dword 32		R0 D	R0 D	
			Long 32		R0 L	R0 L	
			Float 32		R0 F	R0 F	
Extended File register	ZR	524287	bit	ZR0.0			
			Int 16		ZR0	ZR0	
			Word 16		ZR0 W	ZR0 W	
			dword 32		ZR0 D	ZR0 D	
			Long 32		ZR0 L	ZR0 L	
			Float 32		ZR0 F	ZR0 F	

4.3. Mitsubishi L Series PLC

Mitsubishi L series PLC data (device) types supported in the Adroit Protocol.

Device	Prefix	Max Range	PLC Data Type	Boolean	Integer	Real	String
Inputs	X	0x7FF	discrete	X 0			
			Int 16		X0	X0	
			Word 16		X0 W	X0 W	
			dword 32		X0 D	X0 D	
			Long 32		X0 L	X0 L	
			Float 32		X0 F	X0 F	
Output relay	Y	0x7FF	discrete	Y 0			
Internal relay	M	8191	discrete	M 0			
Latch relay	L	8191	discrete	L 0			
Annunciator	F	2047	discrete	F 0			
Link relay	B	0xFFF	discrete	B 0			
Special link relay	SB	0x7FF	discrete	SB 0			
Edge relay	V	2047	discrete	V 0			
Data register	D	6143	bit	D0.0			D 0 Sn
			Int 16		D0	D0	
			Word 16		D0 W	D0 W	
			dword 32		D0 D	D0 D	
			Long 32		D0 L	D0 L	
			Float 32		D0 F	D0 F	
Link register	W	0xFFF	Int 16		W 0	W 0	
Timer (current value)	T or TN*	2047	Int 16		TN 0	TN 0	
Retentive timer (current value)	ST	2047	Int 16		ST 0	ST 0	
Counter (current value)	C or CN*	1023	Int 16		CT 0	CN 0	
Special link reg.	SW	0x7FF	Int 16		SW 0	SW 0	
File register	R	32767	bit	R0.0			R 0 Sn
			Int 16		R0	R0	
			Word 16		R0 W	R0 W	
			dword 32		R0 D	R0 D	
			Long 32		R0 L	R0 L	
			Float 32		R0 F	R0 F	
Extended File register	ZR	524287	bit	ZR0.0			
			Int 16		ZR0	ZR0	
			Word 16		ZR0 W	ZR0 W	
			dword 32		ZR0 D	ZR0 D	
			Long 32		ZR0 L	ZR0 L	
			Float 32		ZR0 F	ZR0 F	

4.4. Mitsubishi iQ-R Series PLC

Mitsubishi R series PLC data (device) types supported in the Adroit Protocol.

Device	Prefix	Max Range	PLC Data Type	Boolean	Integer	Real	String
Inputs	X	0x2FFF	discrete	X 0			
			Int 16		X0	X0	
			Word 16		X0 W	X0 W	
			dword 32		X0 D	X0 D	
			Long 32		X0 L	X0 L	
			Float 32		X0 F	X0 F	
Output relay	Y	0x2FFF	discrete	Y 0			
Internal relay	M	12287	discrete	M 0			
Latch relay	L	8191	discrete	L 0			
Annunciator	F	2047	discrete	F 0			
Link relay	B	0x1FFF	discrete	B 0			
Special link relay	SB	0x7FF	discrete	SB 0			
Edge relay	V	2047	discrete	V 0			
Data register	D	18431	bit	D0.0			D 0 Sn
		18431	Int 16		D0	D0	
			Word 16		D0 W	D0 W	
			dword 32		D0 D	D0 D	
			Long 32		D0 L	D0 L	
			Float 32		D0 F	D0 F	
Link register	W	0x1FFF	Int 16		W 0	W 0	
Timer (current value)	T or TN*	1023	Int 16		TN 0	TN 0	
Retentive timer (current value)	ST	0	Int 16		ST 0	ST 0	
Counter (current value)	C or CN*	511	Int 16		CT 0	CN 0	
Special link reg.	SW	0x7FF	Int 16		SW 0	SW 0	
File register	R	32767	bit	R0.0			R 0 Sn
		32767	Int 16		R0	R0	
			Word 16		R0 W	R0 W	
			dword 32		R0 D	R0 D	
			Long 32		R0 L	R0 L	
			Float 32		R0 F	R0 F	
Extended File register	ZR	4184063	bit	ZR0.0			
		4184063	Int 16		ZR0	ZR0	
			Word 16		ZR0 W	ZR0 W	
			dword 32		ZR0 D	ZR0 D	
			Long 32		ZR0 L	ZR0 L	
			Float 32		ZR0 F	ZR0 F	

4.5. Mitsubishi A Series PLC

Mitsubishi A series PLC data (device) types supported.

Device	Prefix	Max Range	Notation	PLC Data Type	Boolean	Integer	Real	String
Inputs	X	7FF	HEX	discrete	X 0			
				Int 16		X0	X0	
				Word 16		X0 W	X0 W	
				dword 32		X0 D	X0 D	
				Long 32		X0 L	X0 L	
				Float 32		X0 F	X0 F	
Output relay	Y	7FF	HEX	discrete	Y 0			
Internal relay	M	9255	DEC	discrete	M 0			
Latch relay	L	9255	DEC	discrete	L 0			
Annunciator	F	2047	DEC	discrete	F 0			
Link relay	B	4095	HEX	discrete	B 0			
Special link relay	SB	0	HEX	discrete	SB 0			
Edge relay	V	0	DEC	discrete	V 0			
Data register	D	9255	DEC	bit	D0.0			D 0 Sn
				Int 16		D0	D0	
				Word 16		D0 W	D0 W	
				dword 32		D0 D	D0 D	
				Long 32		D0 L	D0 L	
				Float 32		D0 F	D0 F	
Link register	W	FFF	HEX	Int 16		W 0	W 0	
Timer (current value)	T or TN*	2047	DEC	Int 16		TN 0	TN 0	
Retentive timer (current value)	ST	0	DEC	Int 16		ST 0	ST 0	
Counter (current value)	C or CN*	1023	DEC	Int 16		CT 0	CN 0	
Special link reg.	SW	0	HEX	Int 16		SW 0	SW 0	
File register	R	8191	DEC	bit	R0.0			R 0 Sn
				Int 16		R0	R0	
				Word 16		R0 W	R0 W	
				dword 32		R0 D	R0 D	
				Long 32		R0 L	R0 L	
				Float 32		R0 F	R0 F	
Extended File register	ZR	0	HEX	bit	ZR0.0			
				Int 16		ZR0	ZR0	
				Word 16		ZR0 W	ZR0 W	
				dword 32		ZR0 D	ZR0 D	
				Long 32		ZR0 L	ZR0 L	
				Float 32		ZR0 F	ZR0 F	

4.6. Mitsubishi iQ-F Series PLC

Mitsubishi FX series PLC data (device) types supported.

Device	Prefix	Max Range	Notation	PLC Data Type	Boolean	Integer	Real	String
Inputs	X	337	OCT	discrete	X 0			
				Int 16		X0	X0	
				Word 16		X0 W	X0 W	
				dword 32		X0 D	X0 D	
				Long 32		X0 L	X0 L	
				Float 32		X0 F	X0 F	
Output relay	Y	337	OCT	discrete	Y 0			
Internal relay	M	8255	DEC	discrete	M 0			
Latch relay	L	0	DEC	discrete	L 0			
Annunciator	F	0	DEC	discrete	F 0			
Link relay	B	0	HEX	discrete	B 0			
Special link relay	SB	0	HEX	discrete	SB 0			
Edge relay	V	0	DEC	discrete	V 0			
Data register	D	8255	DEC	bit	D0.0			D 0 Sn
				Int 16		D0	D0	
				Word 16		D0 W	D0 W	
				dword 32		D0 D	D0 D	
				Long 32		D0 L	D0 L	
				Float 32		D0 F	D0 F	
Link register	W	0	OCT	Int 16		W 0	W 0	
Timer (current value)	T or TN*	255	DEC	Int 16		TN 0	TN 0	
Retentive timer (current value)	ST	0	DEC	Int 16		ST 0	ST 0	
Counter (current value)	C or CN*	255	DEC	Int 16		CT 0	CN 0	
Special link reg.	SW	0	OCT	Int 16		SW 0	SW 0	
File register	R	0	DEC	bit	R0.0			R 0 Sn
				Int 16		R0	R0	
				Word 16		R0 W	R0 W	
				dword 32		R0 D	R0 D	
				Long 32		R0 L	R0 L	
				Float 32		R0 F	R0 F	
Extended File register	ZR	0	OCT	bit	ZR0.0			
				Int 16		ZR0	ZR0	
				Word 16		ZR0 W	ZR0 W	
				dword 32		ZR0 D	ZR0 D	
				Long 32		ZR0 L	ZR0 L	
				Float 32		ZR0 F	ZR0 F	



PLEASE NOTE

If your PLC's memory range is smaller than the range supported by the driver, it is possible to configure the driver to scan non-existent I/O addresses. Since this can cause unpredictable results please ensure that all scanned I/O are valid for your controller's current configuration.

* Timer/Counter current values are supported, but the contact and coil values are not scanned. T and C prefixes are promoted to TN and CN respectively.

Adroit string slots and ASCII type is only supported to Data/File registers and can be scanned as "D/R 0 Sn" where n is the number of ASCII chars in the string required and which can go up to 79 maximum.

4.7. Omitted

The following I/O types or devices are not supported :

Device	Mnemonic
Function in/outputs	
Special relays	SM, SD
Timer and counter coils and contacts	TS, TC, SS,SC, CS,CC
Step relay	S
Direct in/out	DX,DY

4.8. Non-bit Device Writes

Bit-writes to items that are not supported in the protocol as discrete (bit devices) will be supported using *read-mask-write*, hence they will be slower.

4.9. Bit Devices:

Device	Prefix
Input relay	X
Output relay	Y
Internal relay	M
Latch relay	L
Annunciator	F
Link relay	B
Special link relay	SB
Edge Relay	V

1. Bit devices are always accessed in groups of 16 or more, except when writing to a bit.
2. When addressing Integer and Real slots, it is possible to address 32 bit devices in a grouping. A suffix to the scan-address must also be supplied to indicate the size and presentation.

Supported suffixes for Integer/Real Adroit slots:

Unsigned 16-bit	W
Signed 16-bit (default)	I
Unsigned 32-bit	D
Signed 32-bit	L
Unsigned 64 bit float	F

4.10. Word Devices:

Device	Suffix	Boolean	Integer	Real
Data register	D	D0.0	D0	D0
Link register	W	W0.0	W0	W0
Timer (current value)	T or TN*	none	TN0	TN0
Retentive timer (current value)	SN	none	SN0	SN0
Counter (current value)	C or CN*	none	CN0	CN0
Special link reg.	SW	SW0.0	SW0	SW0
File register	R	R0.0	R0	R0
Ext. File register	ZR	ZR0.0	ZR0	ZR0

1. When accessing a word device in a Boolean or Digital tag, a bit number from 0 through 15 must be supplied.
2. Only when accessing a word device in a Real or Analog tag, does the floating-point presentation become available, (suffix F).
3. Timers and counter types do not support float "F" presentation.

Supported suffixes for Integer/Real Adroit slots:

Unsigned 16-bit	W
Signed 16-bit (default)	I
Unsigned 32-bit	D
Signed 32-bit	L
Unsigned 64 bit float	F

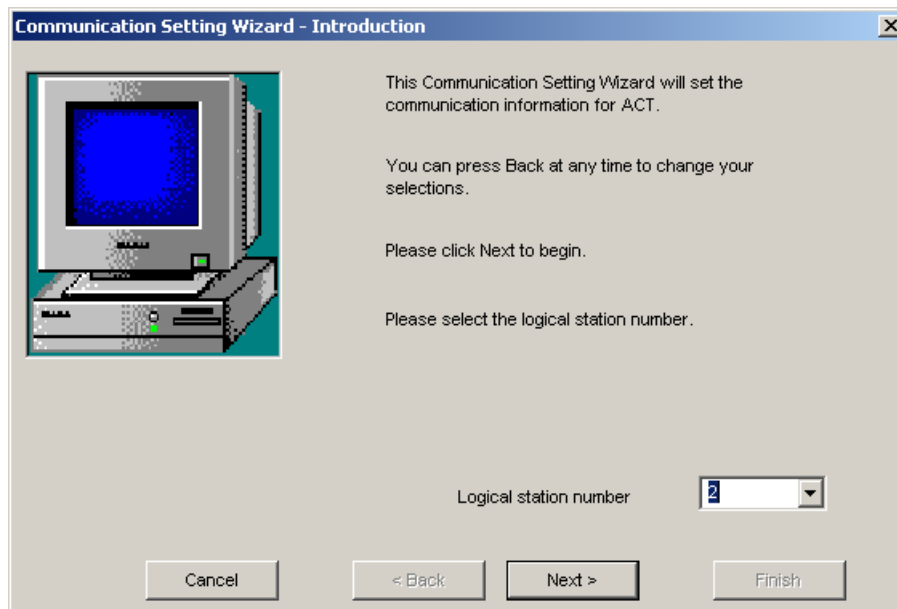
4.11. PLC Software

4.11.1. Installing MX Components

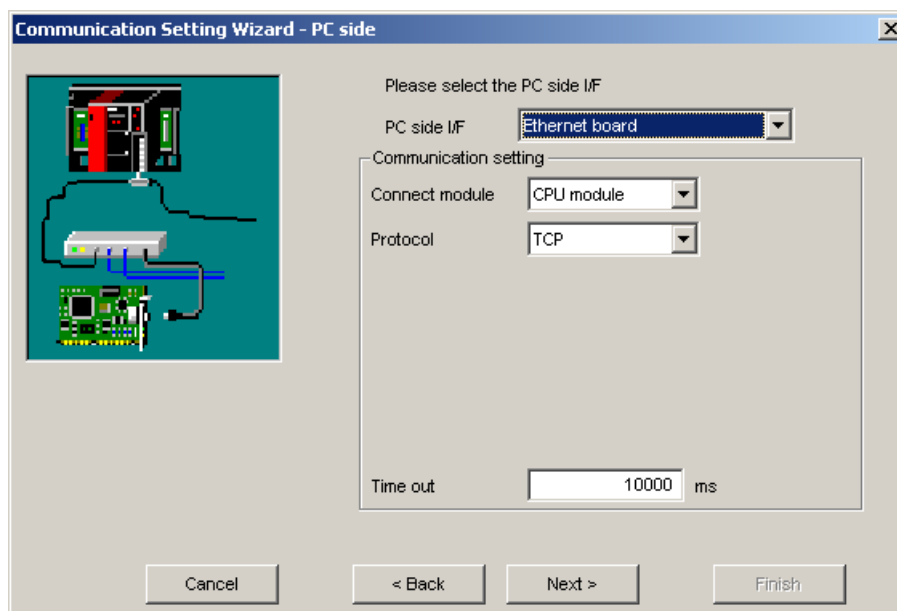
Get the MX Components install from Mitsubishi and run it. It will require a serial key to be entered during the installation so ensure you have a serial key for MX Components. Current version being used is 4.00.

4.11.2. Configuring a new communication channel using the utility.

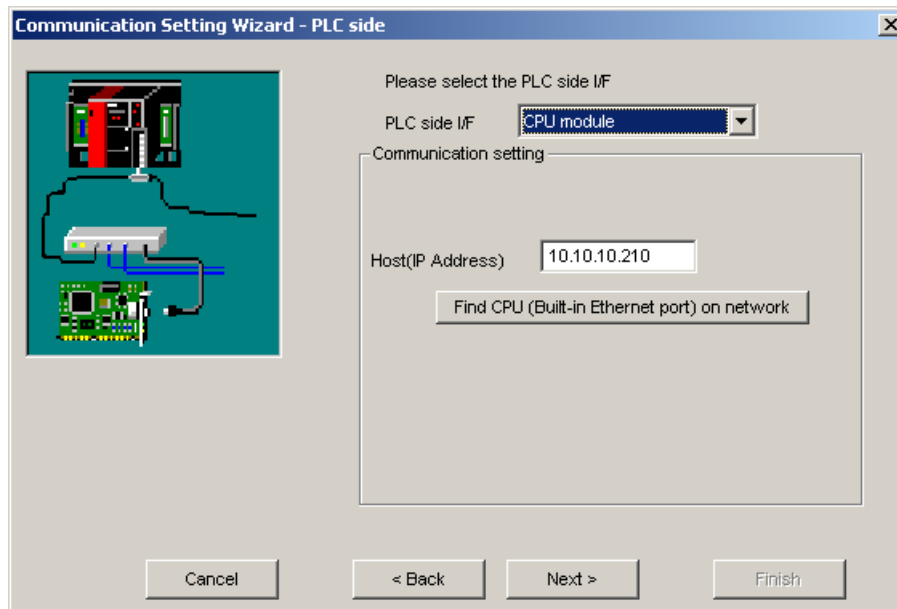
1. Launch Adroit Setup and select the Drivers TAB.
2. Select the MX Components Driver from the list and add a new device.
3. In the device configuration Dialog, press the 'Launch Configuration Wizard' button.
4. You will now have the Mitsubishi communication setup utility open. Press the 'Wizard' button to start the configuration wizard.



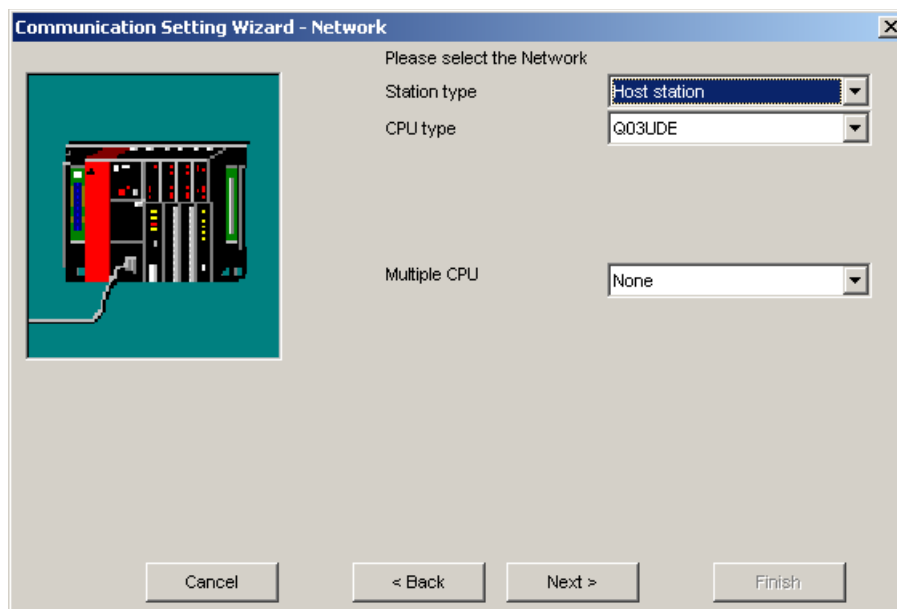
5. From the Logical station number dropdown box you can select an existing configured station to edit or by typing in a new station number you can create a new configuration. Press Next button.



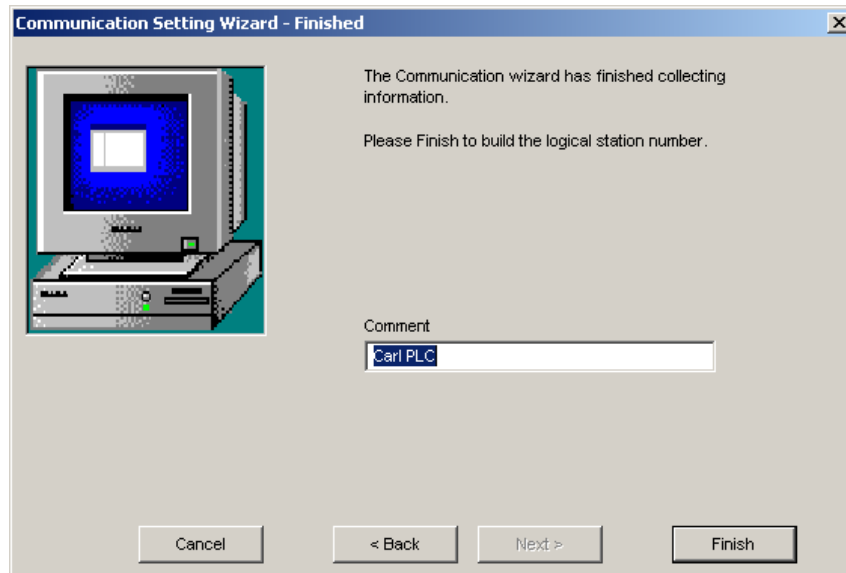
6. Specify the PC side connection interface (Ethernet, serial, simulator, etc.) Then specify which Mitsubishi module on the PLC you want to connect to and what network protocol to use. Lastly specify the connection timeout. Press the Next button.



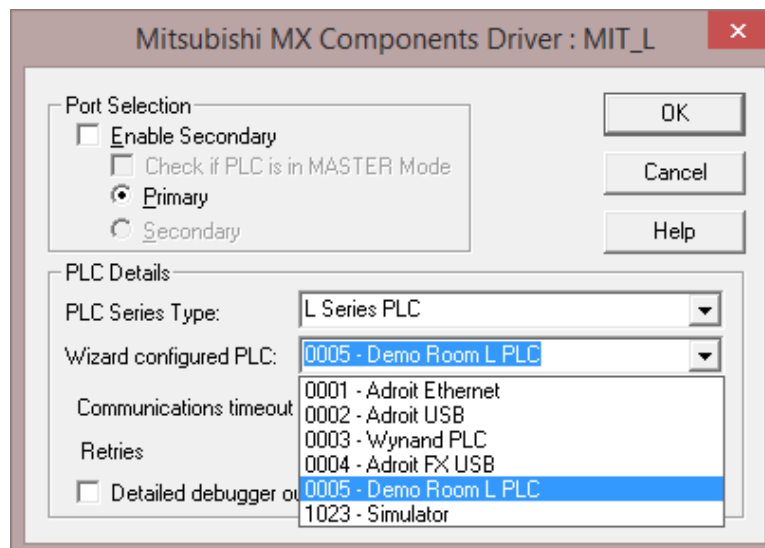
7. Depending on the Connect module selected in Step 6, the PLC side interface can now be selected. Specify the details on how to connect to the PLC like the IP Address if Ethernet was selected in Step 6 or baud rate, parity stop bits, etc. If serial was selected, press the Next button.



8. Now specify if the communication must go to the PLC directly or to another PLC station via some fieldbus network to another PLC. Also specify if the communication must go to the MAIN CPU or if multiple CPUs exist on the rack, to which CPU must communication occur. Press the Next button.



9. Supply your communication configuration with a suitable name. Press the Finish button.
10. Select the Connection Test TAB and press the Test button. Ensure communication is successful.
11. Close the communications setup utility and back on your driver configuration the new configured station will be available for selection in the Wizard configured PLC dropdown box.

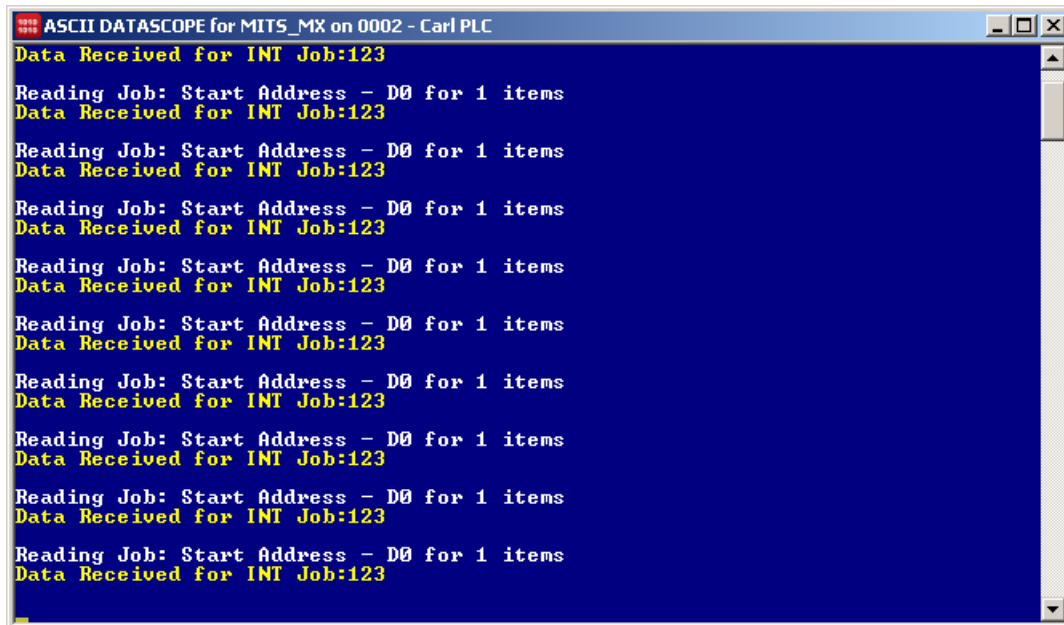


PLEASE NOTE

1. When Starting the Agent Server as a Service the MX Components Driver cannot connect to the GX Simulator 2. For this to work the Agent Server must be started as an application.
2. The MX Components Driver cannot connect to the GX Simulator 2 if the GX Simulator 2 is started after the Agent Server is running. The GX Simulator 2 must be started prior to the Agent Server which will try and connect to it via the MX Components driver.

5. Protocol Definition and Considerations

There is no protocol implemented by the driver as it uses the Mitsubishi ActiveX control to communicate to the PLC's. Below is a screenshot of the datascope for reference.



```
ASCII DATASCOPE for MITS_MX on 0002 - Carl PLC
Data Received for INT Job:123
Reading Job: Start Address - D0 for 1 items
Data Received for INT Job:123
Reading Job: Start Address - D0 for 1 items
Data Received for INT Job:123
Reading Job: Start Address - D0 for 1 items
Data Received for INT Job:123
Reading Job: Start Address - D0 for 1 items
Data Received for INT Job:123
Reading Job: Start Address - D0 for 1 items
Data Received for INT Job:123
Reading Job: Start Address - D0 for 1 items
Data Received for INT Job:123
Reading Job: Start Address - D0 for 1 items
Data Received for INT Job:123
Reading Job: Start Address - D0 for 1 items
Data Received for INT Job:123
Reading Job: Start Address - D0 for 1 items
Data Received for INT Job:123
Reading Job: Start Address - D0 for 1 items
Data Received for INT Job:123
```

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

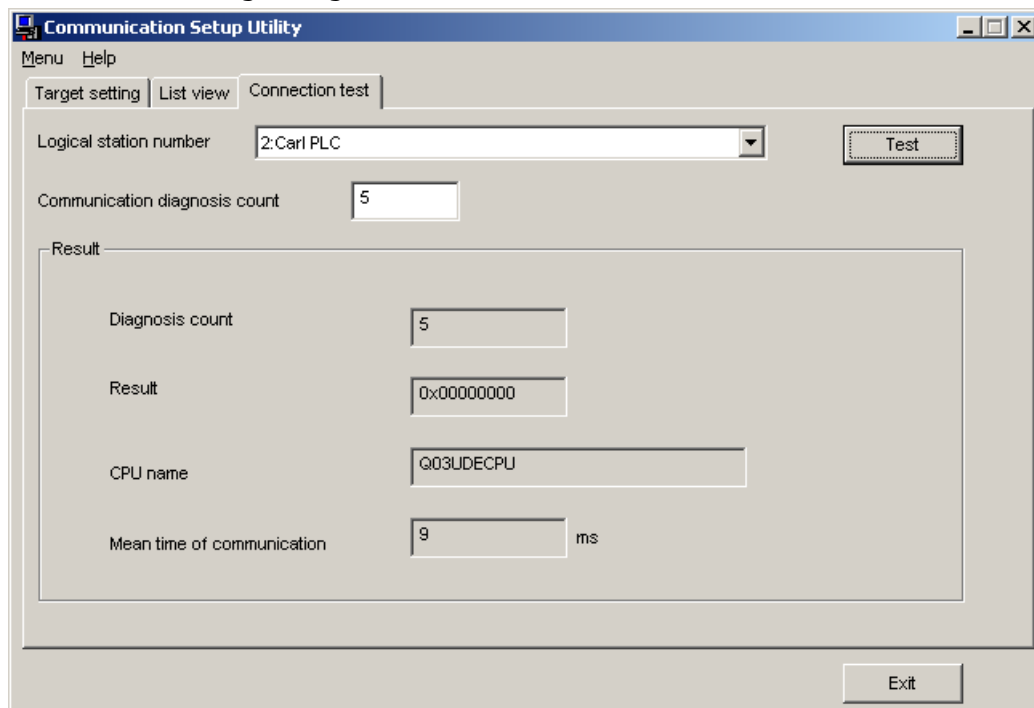
For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

6.3. Troubleshooting

6.3.1. Troubleshooting Steps

1. Ensure you can communicate to the PLC using MX Components by using the communication setup utility to test communications. See image below. Click on the Connection Test TAB and press the test button after selecting configured station.



6.3.2. Error Message Reference

The First 2 bytes of the error message as shown on the Datascope is which PLC Module type send the message, like for example a Q Series CPU. The last 2 bytes in the error message represents the error code as shown in the table below, but only for the Q Series PLC.

Only commonly encountered PLC error codes are listed, for a full reference, see **Melsec Q Programmable Controller User manual (basic) QJ71E71**. The PLC error codes can be seen in the Adroit event log, alarm viewer and driver datascope screens, where a reason description also appears.

HEX Code	Remedial Action
0	No error
4031	The scanned device or I/O does not exist. Verify PLC device range configuration or existence of File register.
C050	Configure the Q71 card for binary and restart the module.
C051 through C055	The number of points to read/write is incorrect, contact Adroit technical support.
C056	The scanned device or I/O does not exist. The starting address + length of the request exceeds the number of points in the PLC. Verify PLC device range configuration.
C057	Requested data length miss-match, contact Adroit technical support.
C058	Requested data length miss-match, contact Adroit technical support.
C059	The command issued is invalid, contact Adroit technical support.
C05B	The requested device does not exist. Verify PLC device configuration.
03E8 through 4FFF	Are generated by the CPU, see CPU error code manual for more detail.

In the event of remedial action failing to rectify a problem, the context of the error message is needed by our team. Please see the Adroit help file on how to capture driver data screens by searching for "to create a protocol dump" (no quotes) in the search tab.

7. Revisions

7.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1	25-Jan-2023	K. Robinson	Changes to format, updated image resources.	D. Jordaan
2.0				

7.2. DLL Revision

Rev.	Date	Changes
15	06-Sep 2021	Now using MX components version 5.
16	13-Jul 2022	Using very fast scanning would cause high CPU load, fixed.

